# Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

## Summary

2026 MS Computer Science graduate building full-stack products with **Python** and **React**, TypeScript and SQL, AI-powered features with LLMs and semantic search, and data-driven iteration from prototype to production.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Advanced NLP, Database Systems, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*

- Restructured fault-tolerant asynchronous **LLM** job orchestration in Python with background workers and failure isolation, processing 1K+ AI jobs for an education platform at 3x throughput without cascading failures.
- Designed multi-tenant **SQL** schemas for 500+ users and tuned indexed queries to cut dashboard latency from 1.2s to 350ms, shipping integrations through CI/CD at 99%+ reliability across 5K+ monthly jobs.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Built a full-stack **AI-powered** product with a **React** frontend and Python Flask APIs, taking upload, rubric, and grading features from design to production for 150+ instructors and 10K+ daily requests.
- Generated rubrics, scores, and written feedback with GPT-4o, Claude, and Qwen-VL, and measured **output quality** at 90%+ grading consistency while cutting instructor feedback turnaround 60% across 10K+ submissions.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer – SecureSync* *India*

- Built object-oriented **Python** and Java modules with Singleton and Factory patterns, covered by 20+ automated unit and integration suites in nightly Jenkins pipelines across Windows, Linux, and macOS.
- Consolidated user activity logs from multiple sources with AWS Glue for **analytics**, boosting pipeline efficiency 68%, and optimized APIs with caching to improve response times 30% for low-bandwidth users.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern – VibrantChem* *India*

- Built a **React and TypeScript** platform with REST microservices and real-time WebSocket updates for customer records, billing, and leads, syncing 2M+ records across multiple locations for business users.
- Used Google Analytics on feature usage and drop-off points to guide **data-driven** product fixes, lifting lead conversions 15% and cutting resource wastage 42% through tested Jenkins CI/CD releases.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*

- Built Python Flask APIs and a React TypeScript dashboard for a **two-sided marketplace** connecting households with service providers, keeping responses under 200ms across 50K+ records on mobile and web.
- Built hourly **Spark** analytics on Databricks over booking events, tracking peak hours, popular services, and daily active users over 50K+ events for internal dashboards and client reporting.

## Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Java, C++, HTML, CSS
**Full Stack:** React, Next.js, Flask, FastAPI, Node.js, REST APIs, WebSockets
**AI & Data:** LLM APIs, Prompt Engineering, RAG, Embeddings, FAISS, Semantic Search, NLP, Spark, Google Analytics
**Databases & Cloud:** PostgreSQL, MySQL, MongoDB, AWS, Docker, Kubernetes
**Practices:** Data Structures, Algorithms, Unit Testing, PyTest, Jest, CI/CD, Git, Code Reviews, Agile, Linux

## Projects

**Related Questions Recommender** | *Python, Embeddings, FAISS, React, A/B Testing*

- Designed a **recommender** that suggests related questions from answer embeddings and reading history, with an A/B test comparing click-through and answer engagement against simpler **topic-based** suggestions for new readers.

**AI Answer Quality Moderation** | *Python, LLM APIs, PostgreSQL, React*

- Designed an **LLM** classifier that flags spam, low-effort, and off-topic answers with reasons, routes borderline cases to human **moderators**, and tracks precision and recall against human reviewer decisions.

**Big-Brain** Live | *LLaMA-2, FAISS, Pinecone, Convex, Shadcn*

- Built a **knowledge** platform with semantic search, Q&A, and summarization over 50K+ documents, plus real-time collaborative conversations, reaching 95% query accuracy and cutting review time 60% for 200+ users.